Tweaking SSINS parameters for optimal outputs

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1 Introduction

Detection of the cosmological 21 cm hydrogen signal from the Epoch of Reionisation (EoR) remains one of the central goals of modern low-frequency radio astronomy. The expected brightness temperature fluctuations associated with the redshifted 21 cm line are extremely faint, typically on the order of approximately 10 mK, making detection highly susceptible to contamination from astrophysical foregrounds, instrumental systematics, and terrestrial Radio Frequency Interference (RFI).

Even weak contamination at amplitudes comparable to the cosmological signal can introduce systematic bias into power spectrum estimation, leading to false detections or incorrect interpretation of the underlying astrophysical processes Nunhokee et al. (2024); Kolopanis et al. (2019); Wilensky et al. (2019). The challenge becomes particularly severe because current-generation interferometers do not directly image the cosmological signal, but instead rely on statistical measurements in Fourier space where residual contamination can mimic cosmological structure.

This work focuses on observations obtained using the Murchison Widefield Array (MWA) Phase III instrument. The upgraded correlator architecture was specifically introduced to mitigate known spectral artefacts associated with previous correlator implementations, where edge and central channels within each 3.24 MHz coarse frequency band required systematic flagging. These regularly missing channels produced harmonic structures in two-dimensional cylindrical power spectra, contaminating measurements within the EoR window.

Although hardware improvements reduce some systematic contamination, software-based RFI mitigation remains essential for ensuring robust cosmological analysis.

2 Problem Statement

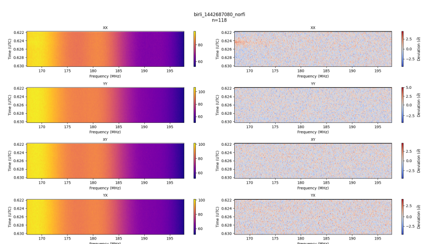
Current MWA preprocessing pipelines rely primarily on two automated flagging algorithms: AOFlagger and the Sky-Subtracted Incoherent Noise Spectra (SSINS) framework.

AOFlagger uses morphological and amplitude-based thresholding methods and performs well for strong RFI contamination (Offringa, 2010). However, previous work has demonstrated that default parameter settings can either fail to detect weak contamination or excessively remove clean data, particularly in low-SNR interferometric observations.

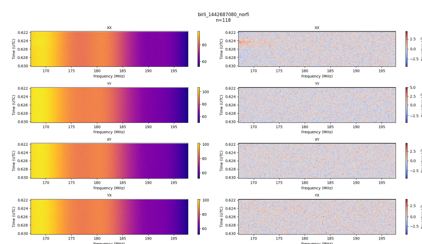
SSINS was introduced to improve statistical identification of contamination by analysing incoherent noise spectra after sky subtraction. While SSINS generally outperforms conventional threshold-based methods, practical analysis of MWA observations reveals inconsistent behaviour. Manual inspection shows three recurring problems:

- Under-flagging, where weak contamination survives the detection stage (Figure 1)
- Over-flagging, where statistically normal fluctuations are incorrectly classified as interference (Figure 2)
- Observation-dependent variability, where a single parameter configuration performs inconsistently across different observations

The central challenge is therefore not simply tuning SSINS hyperparameters, but understanding whether the underlying statistical detection framework itself is sufficient for reliable RFI identification.

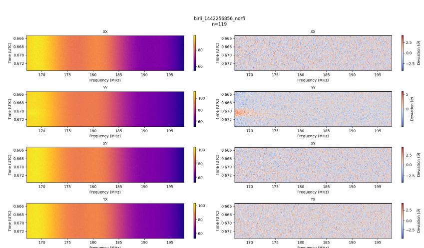


(a) Before SSINS flagging

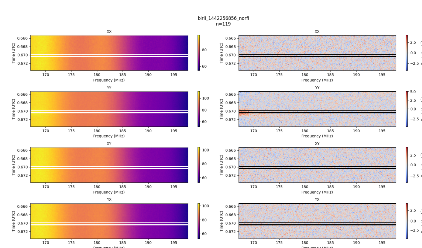


(b) After SSINS flagging

Figure 1: Comparison of visibilities before and after SSINS flagging for observation 1442687080.



(a) Before SSINS flagging



(b) After SSINS flagging

Figure 2: Comparison of visibilities before and after SSINS flagging for observation 1442256856.

3 SSINS Statistical Framework

SSINS detects anomalous visibility behaviour by constructing an incoherent noise spectrum from interferometric visibilities and evaluating deviations from expected thermal noise statistics. At its core, detection is based on the standard score (z-score):

$$z = \frac{x = \nu}{\sigma} \tag{1}$$

where x represents the observed visibility amplitude, ν is the expected mean thermal noise distribution and σ represents the expected standard deviation.

Large positive excursions indicate statistically anomalous behaviour that may correspond to RFI contamination. Unlike conventional amplitude thresholding methods, SSINS searches for deviations relative to the statistical noise model rather than absolute signal strength. This approach is particularly effective for identifying:

- Narrowband persistent transmitters
- Broadband impulsive interference
- Correlator artefacts
- Cable reflections
- Spectrally localised contamination structures

However, the statistical framework introduces several fundamental limitations.

4 Why Z-Score Alone Is Not Sufficient for RFI Detection

Although z-score statistics provide a mathematically convenient anomaly detection framework, they do not fully characterise the complexity of contamination present in radio interferometric observations. A large z-score does not necessarily imply RFI contamination, and conversely, contamination does not always produce large z-score excursions.Several limitations arise.

4.1 Weak Persistent RFI May Not Produce Statistical Outliers

SSINS assumes contamination manifests as statistical deviation from thermal noise behaviour. However, weak continuous interference may remain close to the underlying noise distribution.In such cases, $z < z_{thresh}$ despite contamination being physically present.

This leads to systematic false negatives, where weak interference propagates downstream undetected. This is particularly problematic in EoR experiments because weak contamination can accumulate coherently during long integration times and become visible only in power spectrum space.

4.2 Instrumental Systematics Can Mimic Statistical Outliers

Certain instrumental artefacts produce non-Gaussian deviations similar to RFI contamination such as cable reflections, gain calibration instabilities, correlator errors, bandpass ripple structures and cross-talk between antennas. These systematics can generate large z-score excursions despite not being true external RFI. This causes false positive detections, resulting in unnecessary data loss.

4.3 Temporal Correlation Is Ignored

The z-score framework evaluates each sample independently. However, many contamination sources exhibit temporal coherence, for example satellite passes, aircraft communication bursts and repeating transmission cycles. A single high z-score does not capture temporal evolution. Conversely, weak but temporally correlated contamination may remain statistically insignificant when evaluated independently.

4.4 Spectral Morphology Is Ignored

RFI sources exhibit characteristic spectral signatures.Examples include narrow carrier lines, frequency comb structures, harmonic periodicity and broadband impulsive bursts. Two contamination events may produce identical z-score values while exhibiting completely different spectral behaviour. SSINS does not explicitly model spectral morphology.

5 Hyperparameter Tuning Strategy

The SSINS framework contains several sensitive hyperparameters controlling detection behaviour. The principal parameters investigated are:

1. Detection Threshold: Controls the z-score cutoff for anomaly detection.
2. Narrowband Detection Parameter: Controls sensitivity to spectrally localised contamination
3. Temporal-Spectral Aggregation: Controls smoothing across time-frequency structure.

6 Results

Systematic parameter exploration reveals that no single hyperparameter configuration performs optimally across all observations.Three behavioural regimes emerge.

6.1 Aggressive Flagging Regime

For lower detection thresholds, typically below 4σ, SSINS exhibits highly sensitive detection behaviour. In this regime, strong and moderate RFI structures are effectively removed; however, a substantial fraction of statistically consistent visibilities are incorrectly flagged. This over-flagging results in unnecessary loss of scientifically useful data and can negatively affect downstream calibration by reducing available visibility coverage. Observed characteristics include:

- High overall fraction of flagged data
- Effective suppression of strong narrowband contamination
- Increased false positive detections
- Loss of astrophysical signal information

Although contamination removal is improved, the aggressive threshold introduces a clear trade-off between RFI suppression and signal preservation.

6.2 Conservative Flagging Regime

At higher threshold values, typically above 7σ, SSINS becomes significantly less sensitive to contamination. In this regime, only the most statistically significant outliers are identified, resulting in lower overall flagging fractions and improved data retention.

However, visual inspection of post-flagging diagnostic plots reveals that weak contamination structures frequently remain undetected.This regime is characterised by:

Lower fraction of flagged visibilities

- Preservation of larger data volume
- Weak persistent RFI remaining in the dataset
- Residual contamination propagating into later pipeline stages

Although fewer false positives occur, undetected contamination can introduce systematic bias into subsequent power spectrum estimation.

6.3 Intermediate Detection Regime

Threshold values near the baseline configuration of approximately 5σ provide the most balanced overall performance. In this regime, a significant fraction of contamination structures are successfully removed while maintaining acceptable data retention. Observed improvements include:

- Reduction in coherent contamination structures in time-frequency space
- Improved stability during calibration

- Smoother downstream power spectrum behaviour
- Reduced excessive loss of clean visibility data

However, despite improved performance relative to both aggressive and conservative parameter regimes, residual contamination structures remain visible in several observations.

7 Observation-Dependent Variability

A key result of this study is that identical SSINS parameter configurations behave inconsistently across different observations. Some observations required more aggressive thresholding to suppress strong contamination, while others exhibited excessive flagging under the same configuration. This variability indicates that contamination characteristics are strongly observation-dependent and that static parameter selection cannot reliably generalise across an entire dataset.

Representative analysis of multiple observations showed:

- Certain observations remained under-flagged despite moderate threshold values
- Other observations experienced significant over-flagging under identical conditions
- No globally optimal parameter configuration was identified across all tested observations

This demonstrates that fixed hyperparameter selection is fundamentally limited when applied to heterogeneous observational datasets.

8 Limitations of z-Score-Based Detection

An important result emerging from this analysis is that optimisation of SSINS hyperparameters alone cannot fully address contamination detection challenges.

Even under near-optimal threshold configurations, several contamination structures persist in the data without producing statistically significant z-score excursions. Conversely, instrumental systematics occasionally produce strong statistical deviations that are incorrectly classified as RFI contamination.

These findings suggest that z-score based anomaly detection is inherently limited because it assumes contamination is directly correlated with statistical deviation from expected thermal noise behaviour. However:

- Weak persistent RFI may remain statistically indistinguishable from background noise
- Instrumental artefacts can generate false positive detections

- Temporal and spectral coherence of contamination are not explicitly modelled

As a result, statistical threshold optimisation alone cannot guarantee robust contamination mitigation.

9 Evaluation Metrics

Performance of SSINS tuning is assessed using the following criteria:

- Flagging Fraction Stability: Ensuring consistent data retention across observations.
- Residual RFI Suppression: Visual and statistical reduction of structured contaminants in z-score space.
- Signal Preservation: Minimal distortion of astrophysical visibilities.
- Downstream Impact: Improvement in calibration stability and power spectrum smoothness.

Evaluation should move beyond simply measuring fraction of flagged data. Recommended metrics include Statistical Metrics, Percentage flagged fraction, false positive rate, false negative rate, spectral Metrics, residual coherent spectral structure, bandpass smoothness, cosmological Metrics, improvement in EoR window cleanliness, reduction of wedge contamination, pipeline Metrics, calibration convergence stability and reduction in harmonic artefacts in power spectrum.

10 Implications for Future RFI Mitigation

The results demonstrate that while SSINS remains an effective first-stage statistical detection tool, reliance solely on z-score thresholding is insufficient for reliable identification of weak or structurally complex contamination.

The findings suggest that future RFI mitigation strategies should move beyond static threshold-based detection and incorporate additional diagnostic approaches such as:

- Adaptive observation-specific threshold selection
- Morphological analysis of time-frequency contamination structures
- Machine learning based anomaly detection frameworks
- Downstream power spectrum feedback mechanisms for iterative re-flagging

These approaches may provide improved discrimination between true RFI contamination and instrumental systematics while preserving scientifically useful signal.

11 Iterative Optimisation Procedure

An iterative tuning framework is adopted to systematically refine SSINS performance:

1. Initial Run: Execute SSINS using baseline parameters (threshold ≈ 5).
2. Diagnostic Evaluation: Inspect z-score waterfall plots and flag distributions.
3. Adjustment Step: Increase threshold if over-flagging is observed. Decrease threshold if residual RFI structures persist.
4. Re-evaluation: Quantify improvements using fraction of flagged data reduction in coherent residual structure and stability of downstream power spectra
5. Convergence Check: Iteration continues until additional parameter changes yield negligible improvement.

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